

1 INTRODUCTION

The concern that more autonomous instruments and machines would render work obsolete has recurred throughout history. From Aristotle speculating on the obsolescence of human labor if instruments could work on their own to Keynes’ famous prediction of technological unemployment,² this concern arises whenever new technologies emerge and has particularly grown with recent advances in artificial intelligence (AI). Growing progress in natural language processing and generative models like OpenAI’s GPT have significantly improved machines’ abilities to mimic many human cognitive capabilities, such as understanding and generating human-like text, making predictions, and performing image and video recognition (Dwivedi *et al.*, 2023). Furthermore, the convergence of AI with robotics signals the emergence of highly autonomous systems capable of executing more complex physical tasks, potentially reshaping the landscape of both cognitive and manual labor.

This paper aims to identify which occupations are most exposed to the replacement due to AI adoption, developing a new Generated Index of Occupational Exposure (GENOE). Our approach uses a novel methodological approach that we call “synthetic AI surveys.” This methodology leverages the extensive knowledge and advanced analytical capabilities of large language models (LLMs) to conduct expert-like assessments, provide estimations, and offer insights on a wide range of areas. We specifically applied this approach to estimate the exposure of various tasks and occupations to recent and anticipated advancements in AI-related technologies. The GENOE index measures the likelihood of job replacement by AI at the occupational level, based on the tasks typically performed in each role. We defined three time horizons for replacement: one year, five years, and 10 years. This accounts for short-term scenarios based on current AI capabilities as well as medium-term and long-term scenarios anticipating future technological advancements.

Our findings reveal that the average occupational exposure to AI is 0.28 within one year, increasing to 0.38 and 0.44 over the five- and 10-year horizons, respectively.³ An exposure score of 0.28 suggests that, on average, occupations have a 28% likelihood of potentially being impacted by AI within the next year. To contextualize these results, we calibrated the GENOE index with labor market data from the United States and Mexico. Results show that 43 million and 16 million jobs are exposed to AI in the United States and Mexico, respectively, over the one-year horizon, increasing to 60 million and 22 million over five years, and 70 million and 26 million over ten years. Extrapolating these findings to Latin America and the Caribbean, we estimate that approximately 84 million jobs are exposed to AI within one year, rising to 114 million in 5 years, and 132 million in 10 years. Assuming this average exposure is applicable to worldwide labor markets, it implies that approximately 980 million, 1.33 billion, and 1.54 billion jobs are exposed to AI over the one-, five-, and 10-year horizons, respectively. It is crucial to note that this does not directly translate

²See Aristotle (350 B.C.E./1998), Politics and Keynes (1930).

³GENOE estimates are available online for use by academics, policymakers, and businesses to inform future research and strategic decision-making GENOE. Data can be found in this [online repository](#).

to job losses or complete occupational obsolescence. Rather, it indicates the proportion of occupations that may undergo substantial changes due to AI integration. The impact on employment will likely be more complex, involving occupational transformations, the emergence of new roles, and potential displacement in some sectors.

Our analysis shows that in the United States and Mexico, women are more vulnerable to AI displacement, partly due to their significant presence in office and administrative jobs. In the United States, low-educated and low-wage workers are particularly exposed, while in Mexico, both low- and medium-skilled workers, especially those in middle-income formal jobs, are more vulnerable, whereas informal jobs remain less exposed. This suggests that AI adoption could exacerbate income inequalities by disproportionately impacting lower- and middle-income workers, echoing, to some extent, findings from previous technological advancements (Autor *et al.*, 2006; Autor and Dorn, 2013; Autor, 2019; Acemoglu and Restrepo, 2022).

To better understand the rationale behind the synthetic AI survey assessments, we conducted follow-up surveys, asking the model to explain the considerations behind each occupation’s estimation. These responses revealed that the model’s rationale focuses on several key factors: current and expected technological capabilities and limitations, the routine and non-routine nature of tasks, ethical considerations, the regulatory environment, and cost-benefit analyses. The models evaluated AI’s current abilities and future potential, emphasized the likelihood of replacing routine tasks, and frequently mentioned ethical considerations and safety concerns, particularly for roles with direct human interaction. Regulatory factors were noted for industries with strict standards, and cost-benefit analyses influenced the economic feasibility of automation.

We conducted several alternative exercises to examine how the GENOE index changes with different methodologies and prompts. These variations included omitting occupation names, excluding ethical considerations, assessing tasks individually, and concentrating on specific AI technologies. Most notably, when we instructed the model to disregard ethical, regulatory, and social considerations and focus purely on AI’s technical abilities, we observed a significant increase in the estimated exposure across occupations. This alternative GENOE index, which we term GENOE-Tech, provides an estimation of AI’s potential impact based solely on its growing technical capabilities.

This paper makes several contributions to the literature on AI’s impact on employment. First, it overcomes some limitations of the existing indexes that measure occupational exposure to AI (Brynjolfsson *et al.*, 2018; Webb, 2019; Felten *et al.*, 2021; Tolan *et al.*, 2021; Pizzinelli *et al.*, 2023; Eloundou *et al.*, 2024). Most existing indexes disaggregate occupations into collections of separate tasks, skills, or abilities. The first restriction of this approach is that it overlooks more complex interrelations between tasks or abilities. For instance, journalists and technical writers share several automatable routine tasks related to writing, editing, and generating content. Yet, routine roles in journalism often interact with non-routine roles like creating engaging narratives and storytelling, evaluating the significance of news, and engaging in investigative reporting, all of which require

human judgment, critical thinking, and persuasion. This interrelation will result in a lower AI exposure of journalism routine tasks than those of technical writers.

Another limitation of this approach is that it does not account for the ethical and social context of different occupations performing similar types of tasks. Consider, for example, judges and credit analysts. Both make decisions based on evaluating extensive information, and both roles can significantly impact individuals' lives. However, the nature and scope of their impact differ in important ways. Judges deliver verdicts that can dramatically affect individuals' well-being, freedom, rights, and long-term societal standing. Their role demands a profound understanding of ethical principles and the ability to balance complex cases. The ethical weight and social impact of their decisions, in addition to requiring human judgment and empathy, make it unlikely that societies would accept these roles being legitimately replaced by AI. Credit analysts, while also impacting individual well-being through their decisions on loan approvals or credit ratings, typically operate within a more narrowly defined financial context. Their decisions can certainly affect a person's economic opportunities and quality of life, but generally do not carry the same weight of personal liberty or broader societal impact as judicial decisions. This comparison highlights the importance of considering not just the technical similarities in task types, but also the broader ethical, social, and institutional contexts in which different occupations operate when assessing the potential for AI integration or replacement.

Our approach corrects the limitations of previous indexes by integrating a more holistic task-based analysis that captures the interrelations and complementarities between tasks within occupations. Unlike previous methods that disaggregate occupations into isolated tasks, skills, or abilities, our AI-generated index considers the entire vector of tasks, reflecting a complete view of occupational contributions in production. To address the second limitation, we provide the model in the surveys with both the name of the occupation and its vector of tasks, allowing it to estimate the overall likelihood of AI replacement without imposing restrictive assumptions. This method allows the model to consider the ethical and social context of each occupation organically rather than through predefined criteria, offering a more holistic evaluation of exposure and replacement. Alternative prompt specifications and follow-up surveys suggest that our method effectively accounts for these considerations.

This second concern is also raised by Pizzinelli *et al.* (2023) and is somewhat addressed by their complementarity-adjusted occupational exposure index (C-AIOE). This measure incorporates factors such as work contexts and job zones to account for the likelihood that AI will complement rather than replace human labor.⁴ Our approach goes further by imposing fewer assumptions

⁴Notwithstanding its advancements over previous indexes, the work contexts and job zones retrieved from the O*NET contain several limitations and assumptions. Firstly, it assumes that higher levels of education and training automatically correlate with better integration of AI, which may not account for the specific skill sets and practical contexts required in each occupation. Additionally, the chosen contexts, such as face-to-face communication, responsibility for others' health, and physical proximity, might not fully capture the ethical reasoning required in certain roles, nor the degree of public resistance to AI replacing certain decision-making roles. Furthermore, as the index is built upon the AIOE, it relies solely on the isolated similarity of abilities to AI developments and thus does not account for the exposure of tasks and their interrelations in the workplace.

regarding the specific skills and contexts required for AI integration. Additionally, the GENOE index considers the complex interplay of tasks within occupations and integrating the ethical and social contexts organically.

Another contribution of our paper is the novel use of large language models to generate synthetic assessments that aim to replicate and extend expert assessments. Many studies rely on expert surveys or interviews to build analytical tools like indexes, including in the labor automation area (Frey and Osborne, 2017; Felten *et al.*, 2021; Tolan *et al.*, 2021). Synthetic AI surveys offer several advantages. They can efficiently process and synthesize vast amounts of information from diverse sources very rapidly, enabling a larger volume of questions to be addressed in a shorter time frame. Additionally, synthetic AI surveys can be easily scaled to include extensive sample sizes or repeated iterations, allowing greater granularity of the collected data.

While recent research has begun to explore the use of AI in workforce analysis, our methodology offers a novel approach in this evolving field. For example, Eloundou *et al.* (2024) combined human expert insights with AI-generated assessments to evaluate AI’s potential for enhancing task efficiency across occupations. Our work is a step forward in this approach by exclusively and systematically using ChatGPT to build the entire index through multiple iterations and varying model parameters.

To validate the accuracy of this methodology, we replicated the well-known exposure index of Felten *et al.* (2021), replacing mTurk surveys with synthetic AI surveys. Our approach successfully replicated Felten’s AI Occupational Exposure Index (AIOE) with a correlation of 0.99, suggesting our methodology is robust and reliable for these types of tasks and providing evidence of another area where AI-driven automation is highly feasible.

The remainder of this paper is organized as follows. After this introduction, Section 2 details our data sources and methodology used to build the GENOE index. Section 3 presents the GENOE index, including its key findings and implications. This section also includes several robustness checks to validate our results and replicates other human-like expert assessment using our novel synthetic survey methodology. Section 4 explores how the index can be used in practical applications, analyzing occupational exposure in the U.S. and Mexico labor market. Finally, Section 5 discusses some considerations and limitations of our research, while Section 6 concludes with implications and directions for future research.

2 DATA AND METHODOLOGY

We draw data from the O*NET database, which contains detailed descriptions and attributes for 1,016 occupations in the U.S. labor market under the O*NET-SOC 2019 classification. Occupational characterizations are built and continuously updated from surveys and input from multiple sources, including job analysts, incumbent workers, and industry experts. O*NET attributes—such as abilities, skills, knowledge, and tasks—are divided into six major domains: Worker Characteristics, Worker Requirements, Experience Requirements, Occupational Requirements, Work-